

Chirayu Salgarkar, E.I.T.

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Skills

Programming: Python, C/C++ , MATLAB/Simulink, TypeScript

Machine Learning & Control: PyTorch, libTorch, TorchScript, reinforcement learning, multi-agent systems, control systems design

Systems & Tools: Linux, Apptainer, Docker, Git, ROS, Gazebo, $\LaTeX$

Experience

AI Research Engineer Intern (Co-op), Stealth Startup – Cambridge, MA May 2026 – present

- Analyzing production multi-agent reinforcement learning (MARL) scheduler for distributed digital-twin computation across two-branch C++/libTorch and Python codebase; delivering technical synthesis to senior engineers on architecture and sim-to-real risk
- Implementing MAPPO, reward shaping, baselines for comparison

Graduate Research Assistant, RIT Safe AI Lab – Rochester, NY Aug 2024 – present

- First-authored work on Wasserstein trust regions for multi-agent coordination (oral, AAAI MAPF Workshop 2025; extended version under review at *Results in Control and Optimization*); co-authored RL framework with formal guarantees through optimal transport, published in *Learning for Dynamics and Control* [↗](#)
- Secured \$34K NSF AWARE-AI grant for measuring nonstationarity in multi-agent RL via novel Wasserstein metrics

Research Engineer Intern, Georgia Tech Research Institute – Warner Robins, GA Oct 2023 – May 2024

- Developed front-end tools to simulate electronic systems for USAF aircrew training using TypeScript, React, Git [↗](#)
- Co-led statewide workshop for CTSO/JROTC instructors on classroom robotics integration methods

Software Engineer Intern, Harris Computer – Atlanta, GA Feb 2023 – Aug 2023

- Front and back-end development for hospital preauthorization workflows using React, SQL

Undergraduate Controls Researcher, Mercer University + Georgia Tech – Macon, GA Oct 2020 – May 2024

- Developed control optimization strategies for safety-critical systems; published work on nonsmooth Control Barrier Functions at *Conference on Decision and Control (CDC)*
- Implemented drone-to-bipedal-robot path-planning and feedback control using novel CBFs in ROS, Gazebo

Education

Rochester Institute of Technology – Ph.D. in Mechanical and Industrial Engineering expected May 2028

- 3.8/4.0 GPA – Master of Engineering milestone completed Aug 2026 within PhD program
- NSF International Research Experiences for Students (IRES) Fellow – University West, Sweden, Fall 2026

Mercer University – B.S.E. in Biomedical Engineering and Mathematics May 2024

- Stamps Scholar – full-ride scholarship for top entering students, graduated *cum laude*

Leadership

Poll Manager for Gwinnett County Elections, AWARE-AI Trainee Council member (NSF-funded), former varsity collegiate debater (NPDA), VP of RIT Tango Club